

**Degree: B.Tech., Stream: ALL (Group- A)**

**Term - I, First Semester Examination, August - 2025**

**Subject Code: ESCME102A**

**Subject Name: Engineering Mechanics-Principles**

**Full Marks: 30**

**Duration: 1 Hour**

**Part - A**

**Attempt any 5 questions**

**Each question carries 2 marks (2 × 5)**

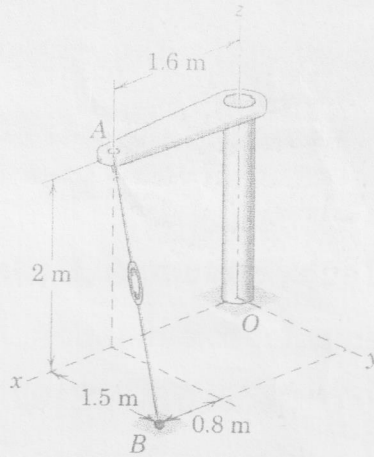
1. State and explain Lami's theorem for a 3-force member in equilibrium 2
2. Draw and explain the parallelogram Law of Vector Addition 2
3. A force  $\vec{F} = 3\hat{i} + 2\hat{j} + 1\hat{k}$  N acts at a point located at position vector  $\vec{r} = 2\hat{i} - 1\hat{j} + 4\hat{k}$  m. Find the moment  $\vec{M}$  of the force about the origin. 2
4. What are the necessary conditions for a body subjected to two forces to be in equilibrium? Explain with reasoning. 2
5. A light rod is in equilibrium under the action of three coplanar forces. Two of the forces are known, and their lines of action intersect. Can you comment anything about the line of action of the 3<sup>rd</sup> force? If yes, state your reasoning behind it. 2
6. If a horizontal force  $F$  is acting on a body of mass  $m$  resting on a flat rough surface (whose coefficient of static friction is  $\mu_s$ ) and it is given that  $F < \mu_s mg$ ; then what is the actual value of friction force ( $F_f$ ) acting on the body? 2

**Part - B**

**Attempt any 2 questions**

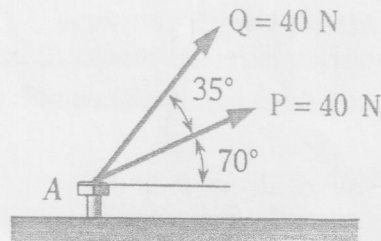
**Each question carries 5 marks (5 × 2)**

7. The tension in cable AB is 2.4 kN. Determine the moment about point O of the cable force acting on point A and the magnitude of this moment. 5



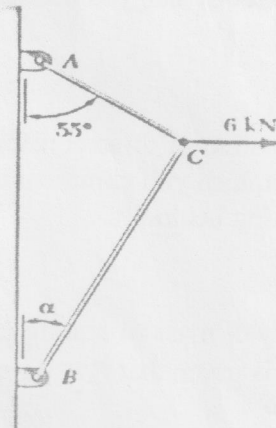
8.

A bolt at point A is being acted upon by two forces, P and Q, as shown in the figure. Determine the magnitude and direction of the resultant force acting on the bolt using the Parallelogram Law.



5

9. Two cables are tied together at C and loaded as shown. If  $\alpha = 30$  degrees, calculate the tension in both cables AC and BC, considering static equilibrium condition.



5

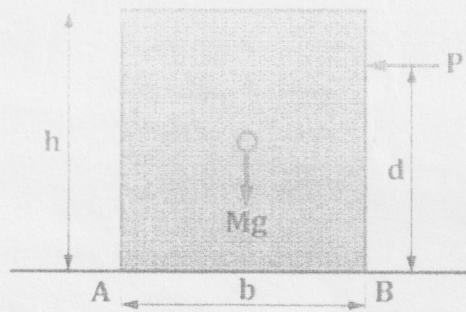
### Part - C

Attempt any 1 question

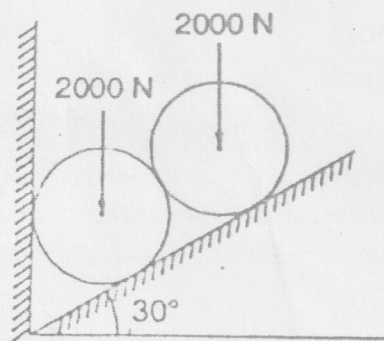
Each question carries 10 marks ( $10 \times 1$ )

10. A rectangular block of mass  $M = 10$  kg rests on a floor. The coefficient of friction between the block and the floor is  $\mu = 0.3$ . The height of the block is  $h = 0.5$  m, and its width are  $b = 0.2$  m. What is the highest position (d) for applying a horizontal force P that would permit to just move the block without tipping?

10



11. Two identical rollers, each of weight 2000N, are supported by an inclined plane and a vertical wall as shown in the figure below. Determine the reaction at the supports. Assume all the surfaces to be smooth.



10

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